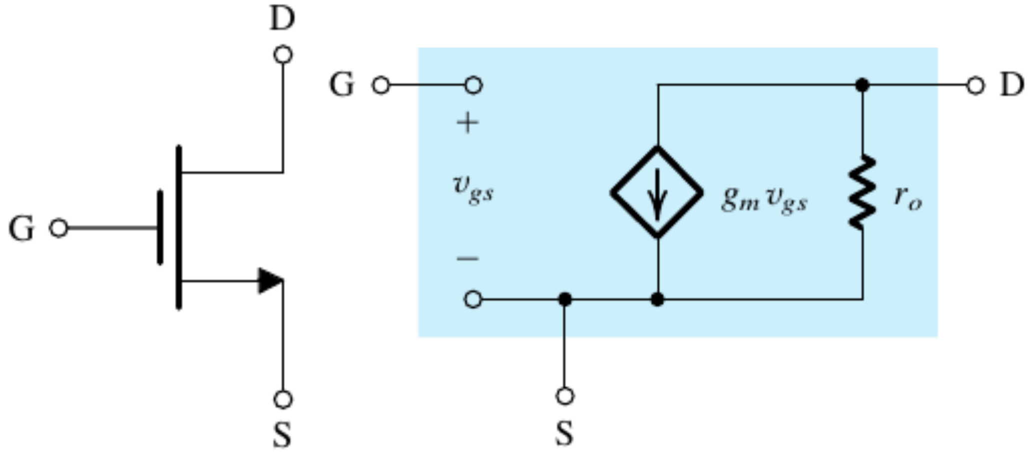
Lecture 7a

EE-215 Electronic Devices and Circuits

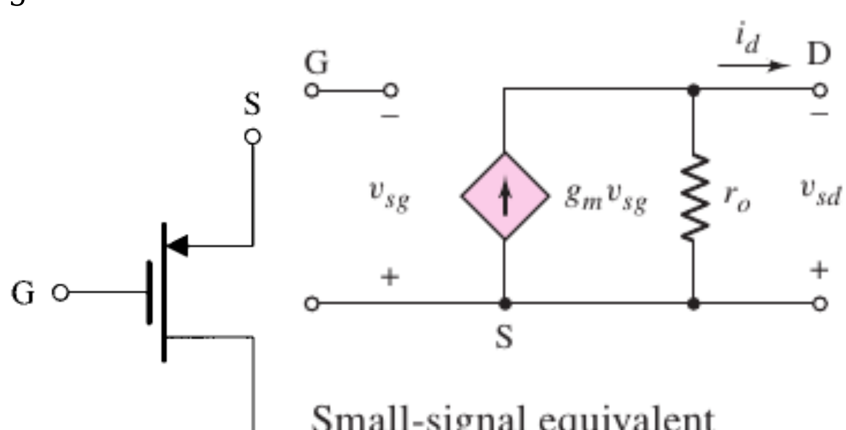
Asst Prof Muhammad Anis Ch

Small-Signal Equivalent Circuit Models

- we had already seen the small-signal equivalent circuit model (hybrid-π model) for an NMOS transistor



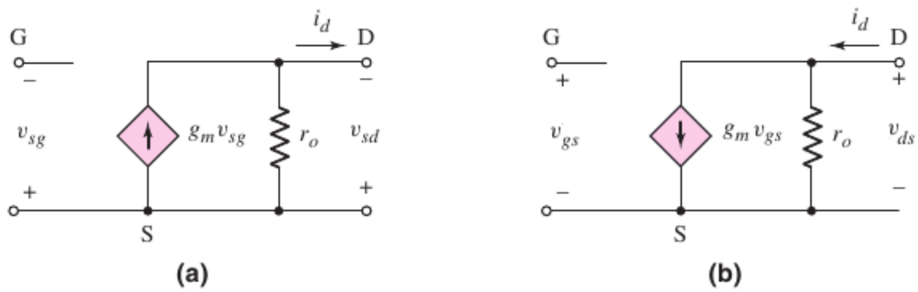
- also for PMOS transistor, the MOSFET acts as a voltage controlled current source.
    - thus the small-signal model for the p-channel MOSFET can be given as



Small-signal equivalent circuit of PMOS transistor

- In the small-signal model of PMOS transistor,
    - if we substitute, $v_{sg} = -v_{gs}$

        - $= \Rightarrow g_m v_{sg} = -g_m v_{gs}$
        - i.e. if the control voltage polarity is reversed, then the dependent current direction is also reversed.
    - also $v_{sd} = -v_{ds}$

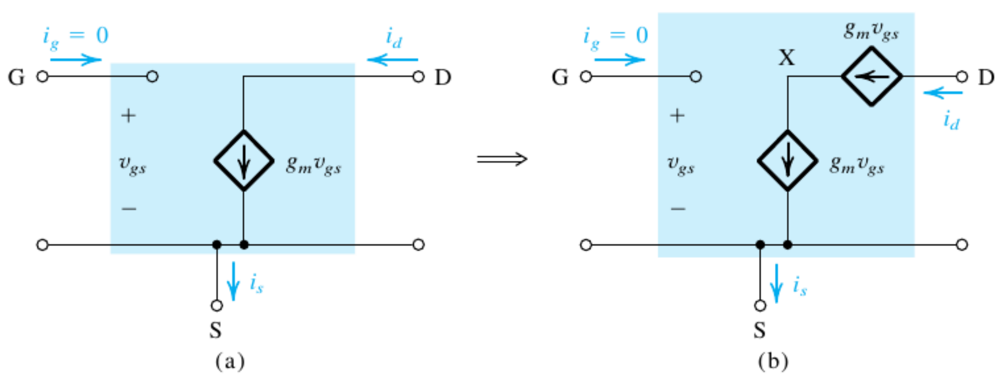


Small signal equivalent circuit of a p-channel MOSFET showing (a) the conventional voltage polarities and current directions and (b) the case when the voltage polarities and current directions are reversed.

- thus the resultant equivalent circuit (fig b) for the PMOS transistor is exactly the same as that of the NMOS transistor
    - we conclude that the small-signal equivalent circuit model (hybrid-π model) is
        - exactly the same for both the NMOS and the PMOS transistors

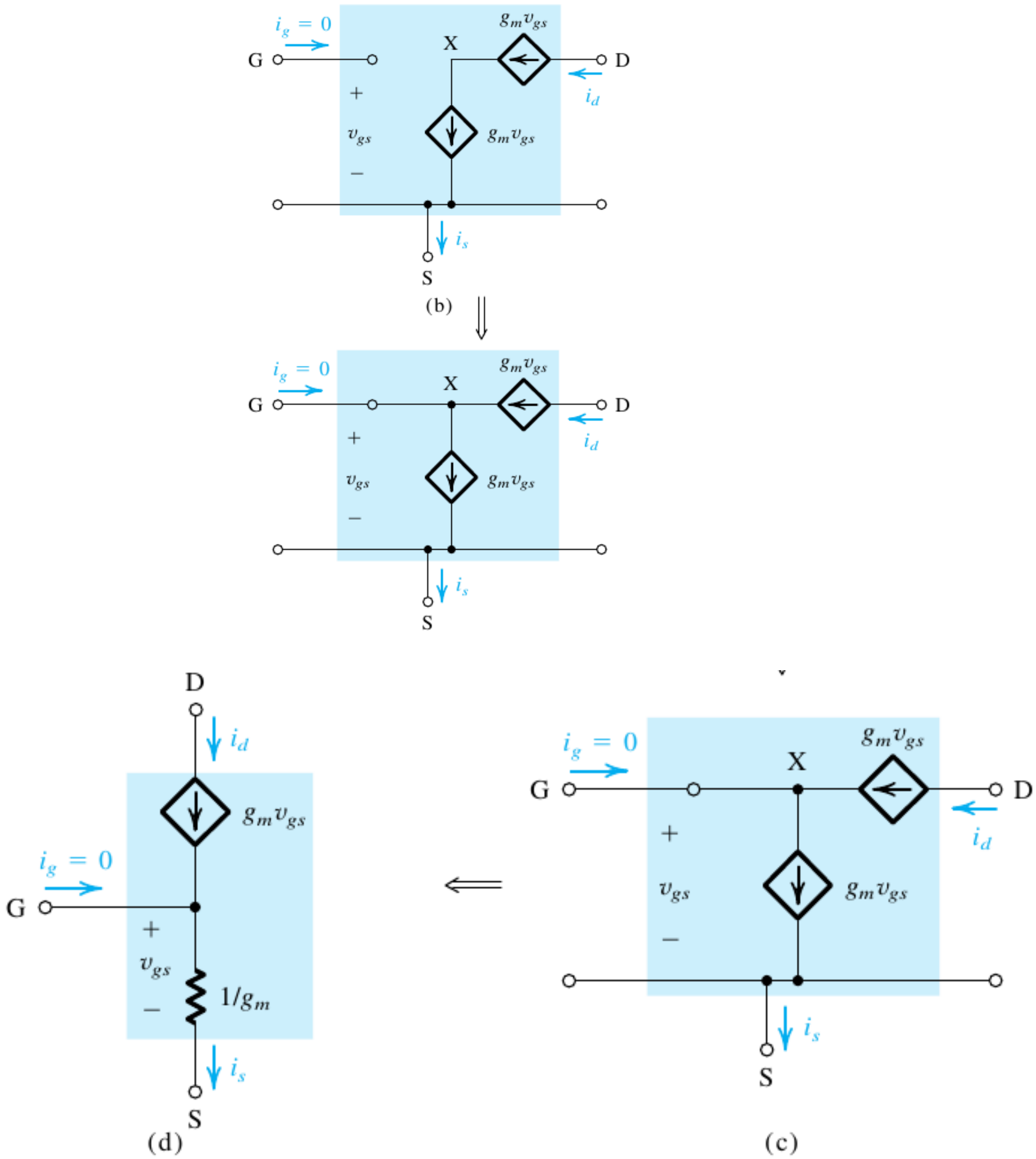
The T Equivalent-Circuit Model

- A simple circuit transformation can lead to an alternative
    - equivalent circuit model for a MOSFET
    - we start with the hybrid-π model



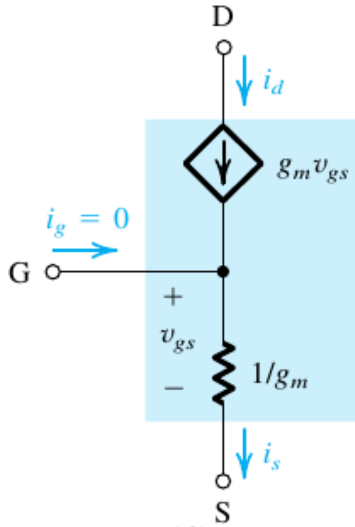
    - here we have placed a 2nd $g_m v_{gs}$ current source in series with the original controlled source.
    - Note that, this new current source doesnot change the terminal currents and is thus allowed
- the new node X, can be connected to the gate.

- As because of KCL, still the gate current $i_g = 0$.
    - so the gate current is still zero (i.e. this connection doesnot change the terminal characteristics)



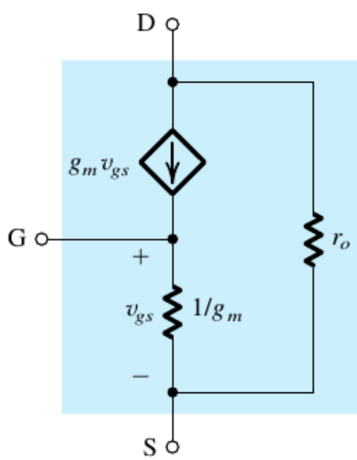
- Note that, here we have a controlled current source $g_m v_{gs}$
    - whose current is controlled by the control voltage v_{gs}
        - where v_{gs} is across itself
    - = ⇒ this current source can be replaced by a resistor of value $\frac{v_{gs}}{g_m v_{gs}} = \frac{1}{g_m}$
- Note that the resistance between the gate and the source,

- looking in to the source terminal is $1/g_m$
  - and looking in to the gate terminal is infinite



(d)

- if the channel-length modulation effect cannot be ignored,
  - we can include r_o in the T-model between the drain and the source terminal



(a)

- Also we can have an alternative representation of the T model,
  - where we can replace the voltage-controlled current source by
    - a current controlled current source

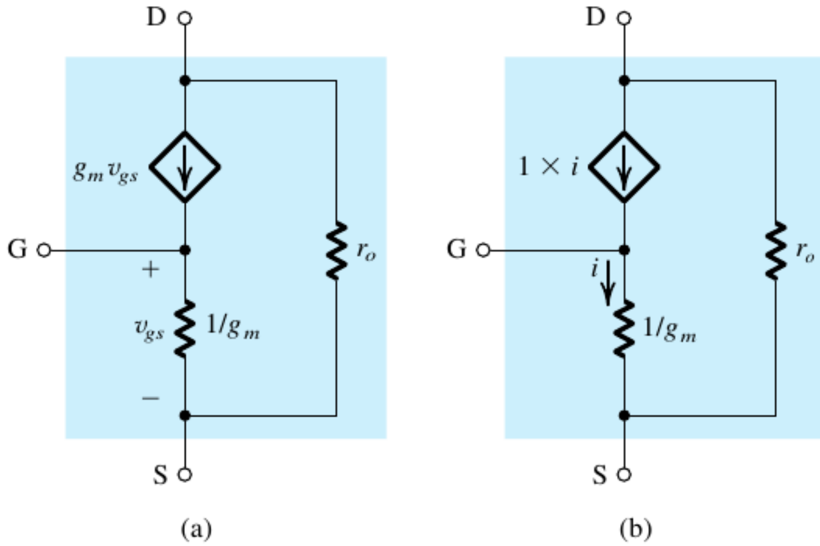


Figure 5.41 **(a)** The T model of the MOSFET augmented with the drain-to-source resistance r_o. **(b)** An alternative representation of the T model.

Small-Signal Operation and Models

Table 5.3 Small-Signal Equivalent-Circuit Models for the MOSFET

Small-Signal Parameters

NMOS transistors

- Transconductance:

$$g_m = \mu_n C_{ox} \frac{W}{L} V_{OV} = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_D} = \frac{2 I_D}{V_{OV}}$$

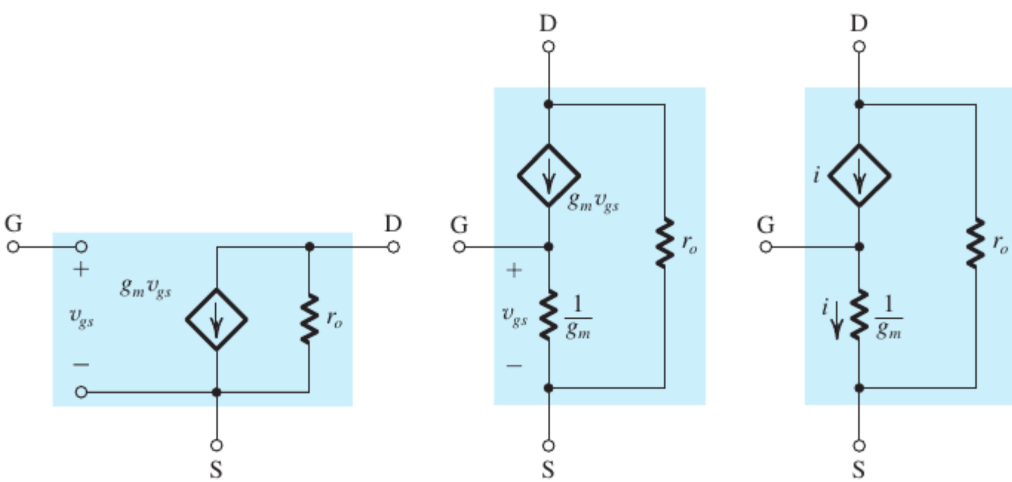
- Output resistance:

$$r_o = V_A / I_D = 1 / \lambda I_D$$

PMOS transistors

Same formulas as for NMOS *except* using $|V_{OV}|$, $|V_A|$, and replacing μ_n with μ_p.

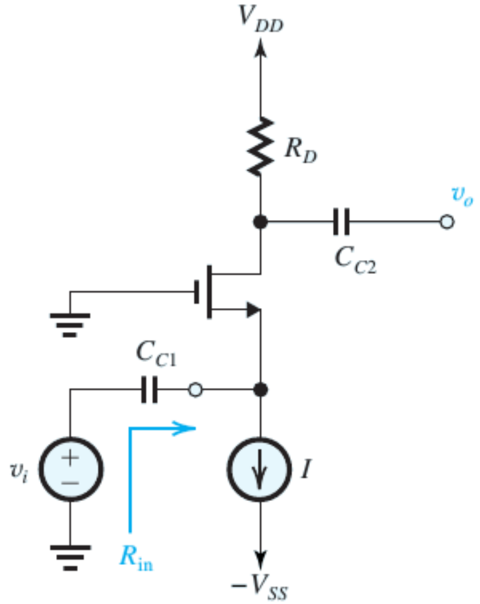
Small-Signal Equivalent Circuit Models



Hybrid-π model T models

Example 5.11

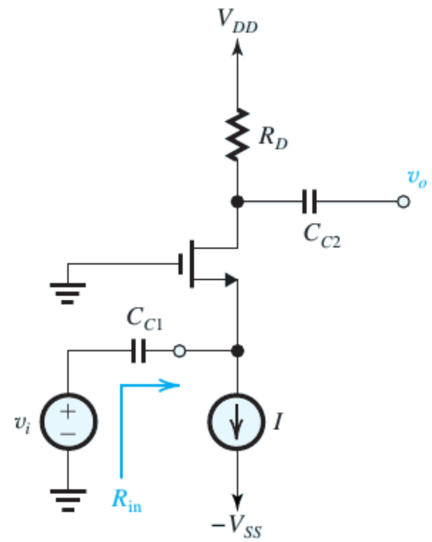
- Figure 5.42(a) shows a MOSFET amplifier biased by a constant-current source I. Assume that the values of I and R_D are such that the MOSFET operates in the saturation region. The input signal v_i is coupled to the source terminal by utilizing a large capacitor C_{C1}. Similarly, the output signal at the drain is taken through a large coupling capacitor C_{C2}. Find the input resistance R_{in} and the voltage gain v_o / v_i . Neglect channel-length modulation.



(a)

Solution:

- here $\lambda = 0$, R_{in} =? , $A_v = \frac{v_o}{v_i}$ =?
  - as an ac source is connected at the source terminal of the MOSFET, it is more convenient to use T-model.
    - thus for ac analysis,
    - suppress dc sources i.e. V_{DD} is replaced by short circuit
      - and I is replaced by an open circuit
    - C_{C1} , C_{C2} are replaced by short circuits
    - MOSFET is replaced by its T-model



(a)

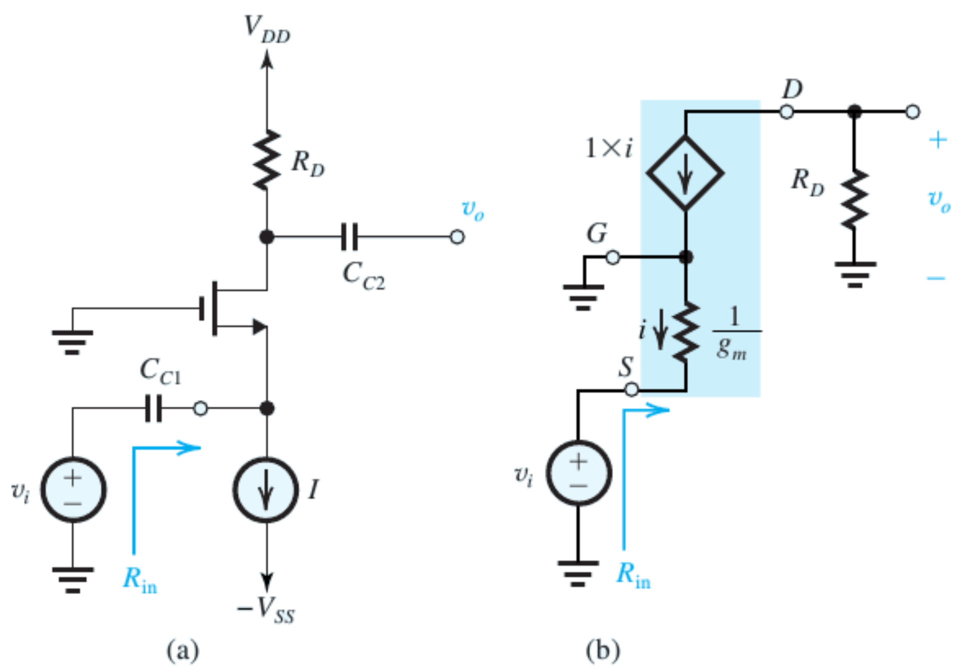
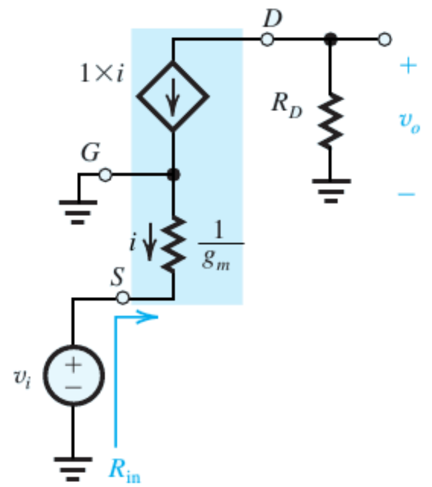


Figure 5.42 (**a**) Amplifier circuit for Example 5.11; (**b**) Small-signal equivalent circuit of the amplifier in (a).

- as $\frac{v_{gs}}{i} = \frac{1}{g_m}$ and $v_{gs} = -v_i$
  - $= \Rightarrow \frac{-v_i}{i} = \frac{1}{g_m}$
    - and $R_{in} = \frac{v_i}{i_i} = \frac{v_i}{-i} = \frac{-v_i}{i} = \frac{1}{g_m}$
    - by ohm's law at R_D
      - $v_o = (-i)R_D$
      - As $i = \frac{v_{gs}}{1/g_m} = g_m v_{gs}$

- $= \Rightarrow v_o = -g_m v_{gs} R_D$
- or $v_o = -g_m\left(-v_i\right)R_D \because v_{gs} = -v_i$
- $v_o = g_m v_i R_D = \Rightarrow A_v = \frac{v_o}{v_i} = g_m R_D$



(b)

Basic MOSFET Amplifier Configurations

Basic MOSFET Amplifier Configurations

- we have already determined that
    - almost-linear amplification can be obtained by
        - biasing the MOSFET at a suitable point Q in
        - its saturation region of operation
        - and by keeping the signal v_{gs} small

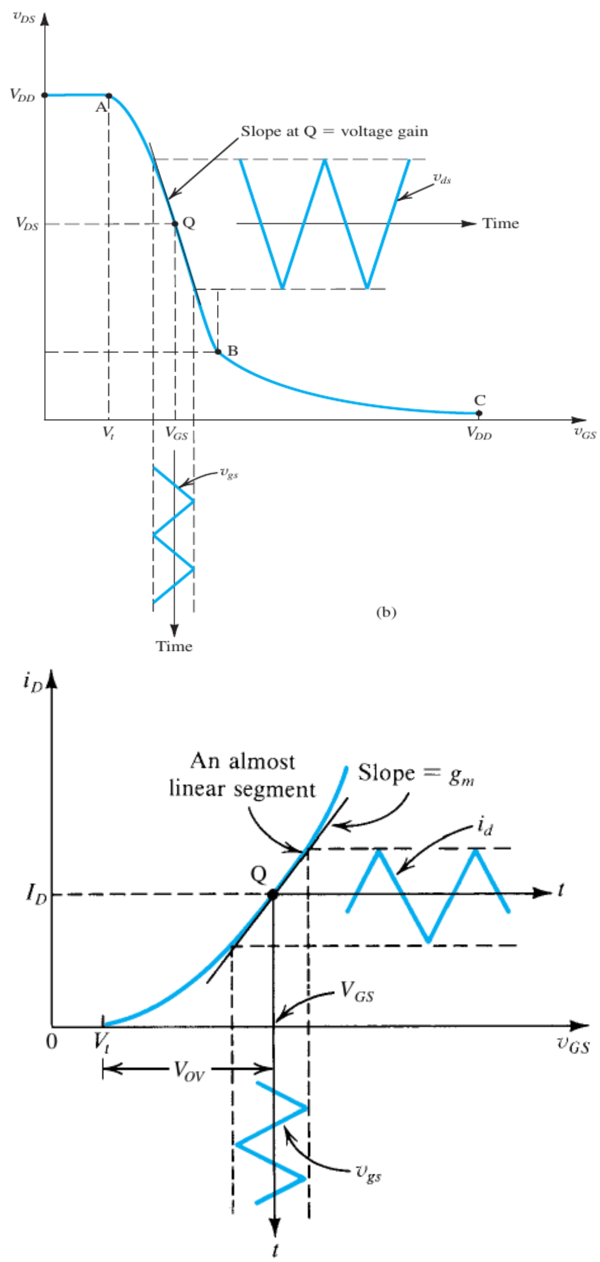


Figure 5.35 Small-signal operation of the MOSFET amplifier.

Basic MOSFET Amplifier Configurations

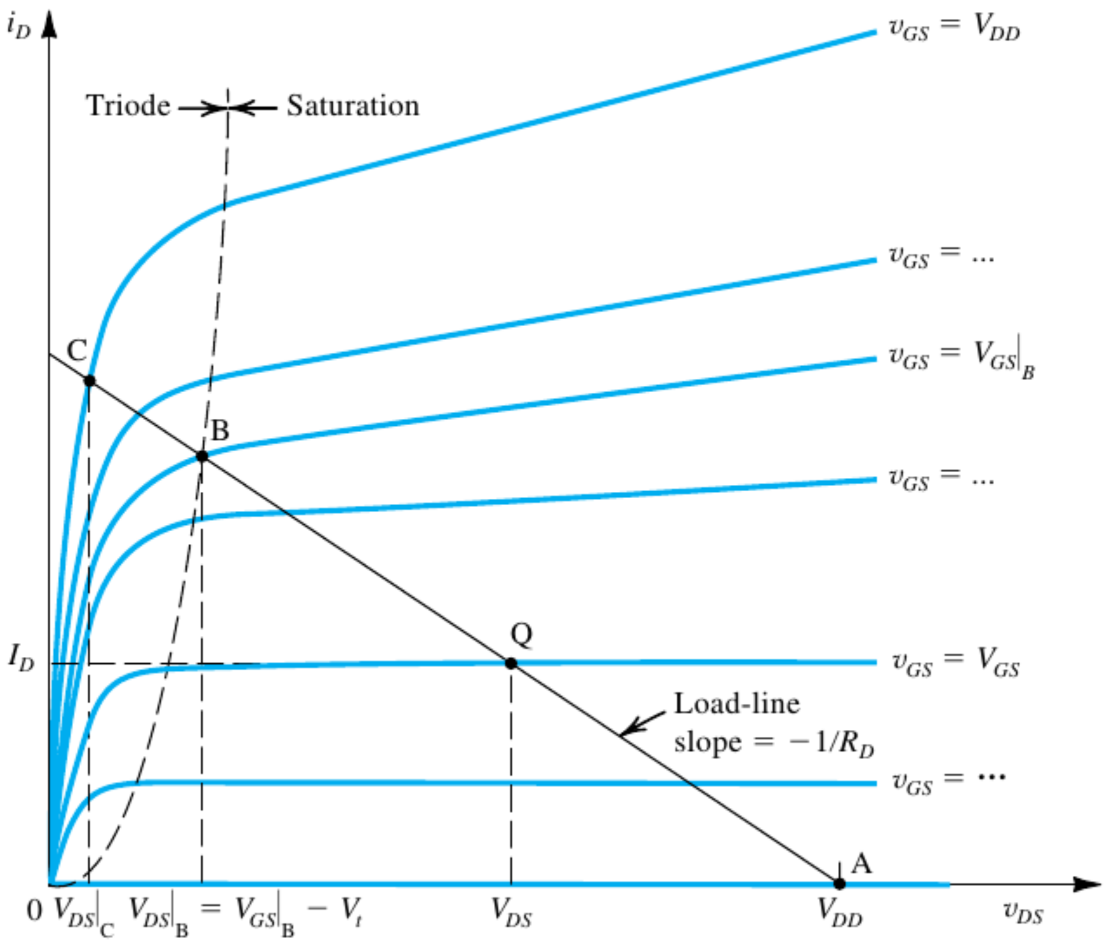


Figure 5.31 Graphical construction to determine the voltage transfer characteristic of the amplifier in Fig. 5.29(a).

- Also when v_{gs} is small, the MOSFET can be replaced by its small-signal circuit model (either hybrid-π model or the T-model)
    - the resultant ac circuit can be used to determine the amplifier parameters like
        - the voltage gain,
        - the input resistance
        - and the output resistance

Table 5.3 Small-Signal Equivalent-Circuit Models for the MOSFET

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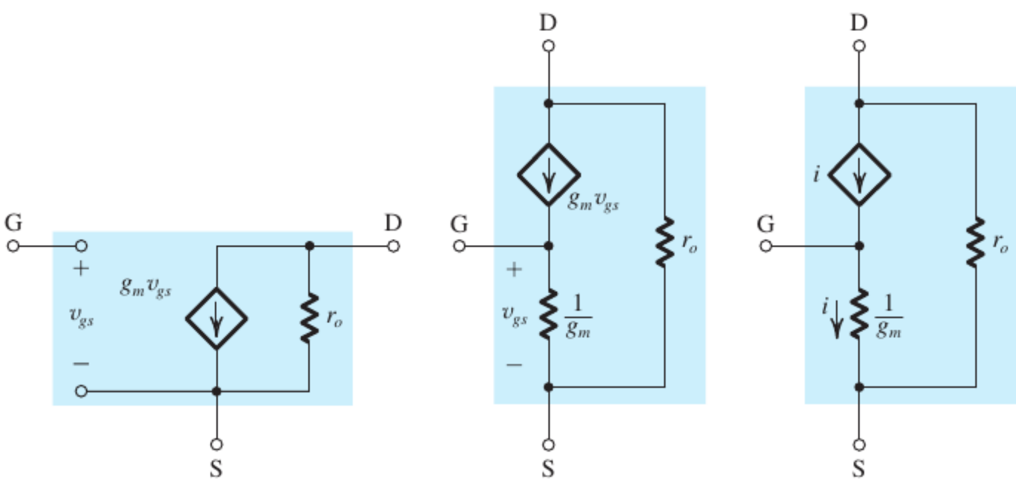
PMOS transistors

Same formulas as for NMOS *except* using $|V_{OV}|$, $|V_A|$, and replacing μ_n with μ_p.

Small-Signal Equivalent Circuit Models



Hybrid-π model

T models

The Three Basic Configurations

- There are three basic configurations for connecting the MOSFET as an amplifier, namely
  - the Common Source (CS) Amplifier
    - the Common Gate (CG) Amplifier
    - The Common Drain (CD) Amplifier (also called Source Follower)
  - Each of these configurations is obtained by
    - connecting one of the three MOSFET terminals to ground
    - thus creating a two-port network with the grounded terminal
    - being common to the input and output ports
  - The resulting three configurations with biasing arrangement ommited are shown in fig

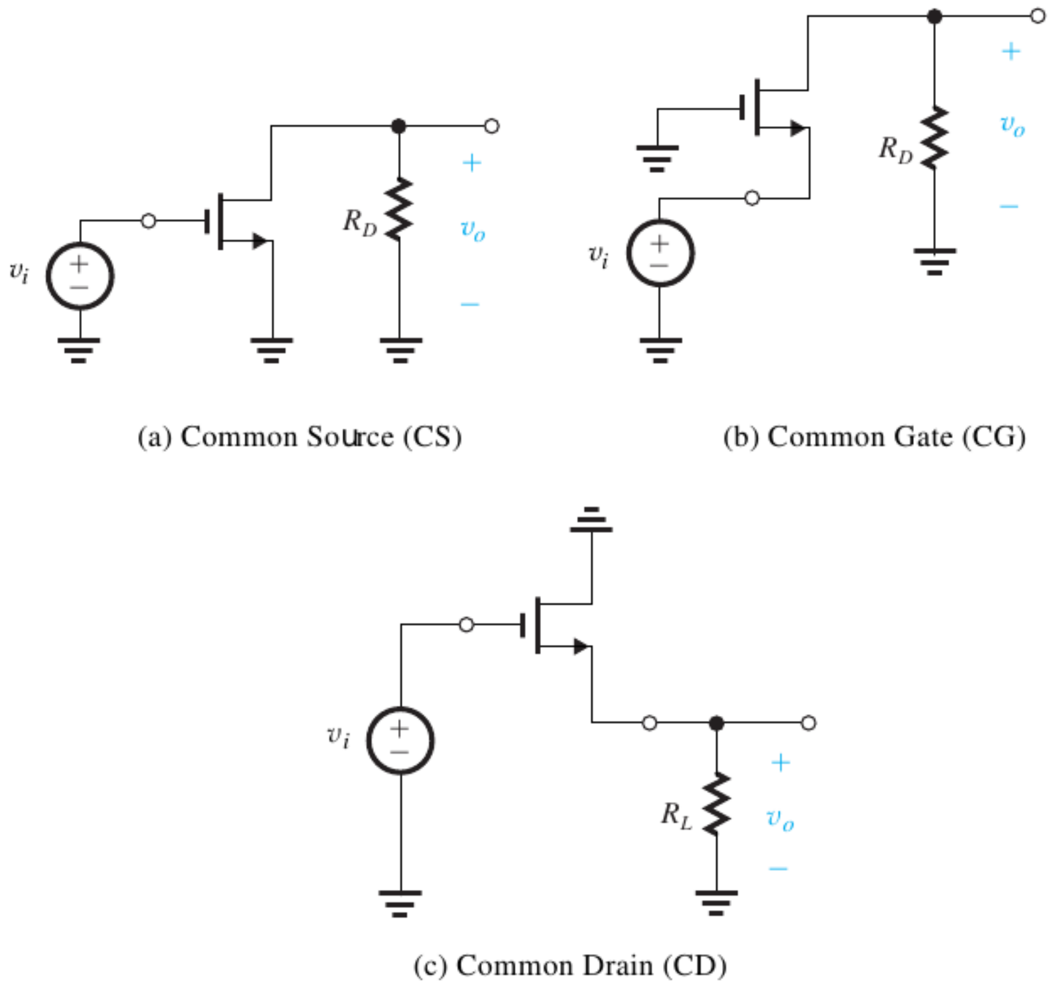
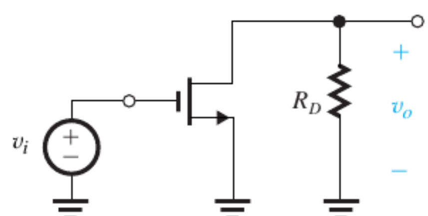


Figure 5.43 The three basic MOSFET amplifier configurations.

Common Source Amplifier

- here the source terminal is connected to ground,
    - the input voltage is applied at the gate (w.r.t ground),
        - the output voltage is taken at the drain (w.r.t ground)

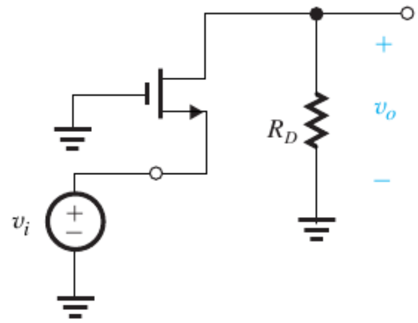


(a) Common Source (CS)

- this configuration is called the grounded source or the common source (CS) amplifier
- Note that dc biasing circuit is not shown in figure

Common Gate Amplifier

- The common gate (CG) or grounded gate amplifier is obtained
    - by connecting the gate to ground,
        - applying the input between the source and ground
            - taking the output v_o between the drain and ground

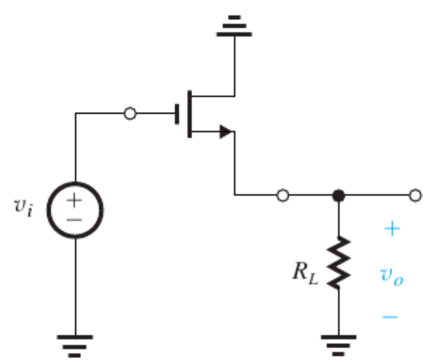


(b) Common Gate (CG)

- Note that dc biasing circuit is not shown in figure

Common Drain Amplifier

- The common drain (CD) or grounded drain amplifier is obtained
    - by connecting the drain terminal to ground,
        - applying the input between the gate and ground
            - taking the output v_o between the source and ground



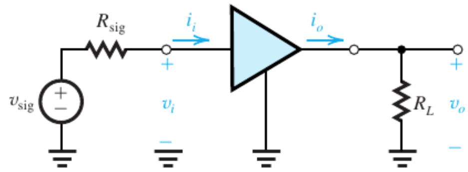
(c) Common Drain (CD)

- Note that dc biasing circuit is not shown in figure
- the Common Drain amplifier is also called Source Follower

Characterizing Amplifiers

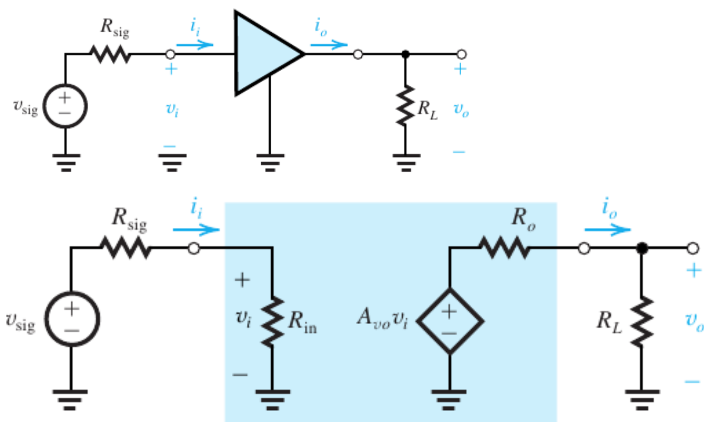
- lets take a look at how to characterize the performance of an amplifier as a circuit building block

- here an amplifier is fed with a signal source
  - having an open-circuit voltage v_{sig} and an internal resistance R_{sig}



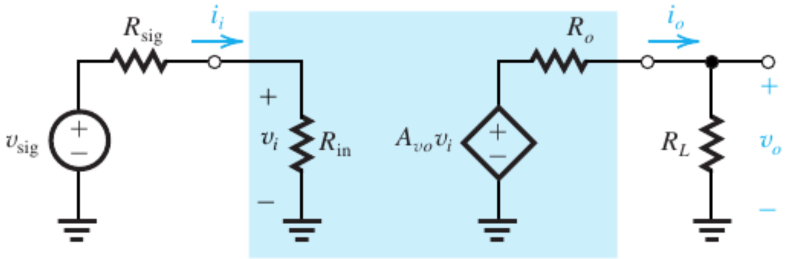
- this voltage source (v_{sig}, R_{sig}) can be an actual signal source
  - or in a cascade amplifier, it can be the Thevenin equivalent of the output circuit of the preceeding amplifier.
- the load resistor R_L is connected at the output terminal of the amplifier
  - this R_L can be an actual load resistor
  - or the input resistance of a succeeding amplifier stage in a cascade amplifier

- The amplfier block can be replaced by its equivalent circuit model



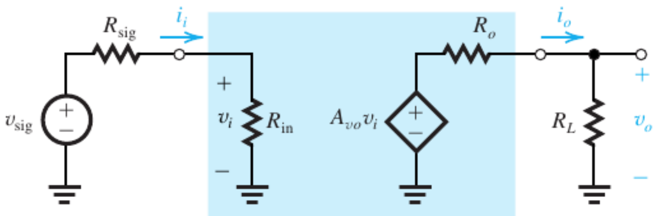
(b)

- the input resistance R_{in}
  - indicate the loading effect of the amplifier input on the signal source
  - and is given as $R_{in} = \frac{v_i}{i_i}$
- by voltage divider rule
  - $v_i = \frac{R_{in}}{R_{in}+R_{sig}} v_{sig}$

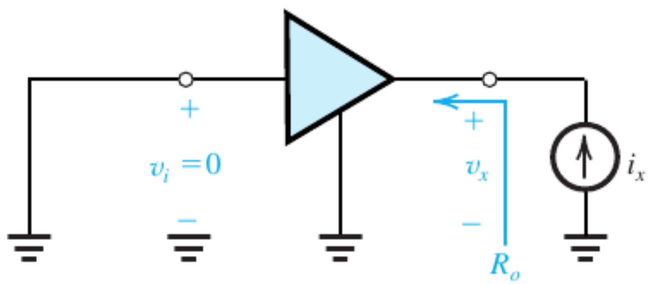


(b)

- $v_i = \frac{R_{in}}{R_{in}+R_{sig}} v_{sig}$
  - A_{vo} is the open-circuit voltage gain and is defined as
    - $A_{vo} = . \frac{v_o}{v_i}\Big|_{R_L=\infty}$

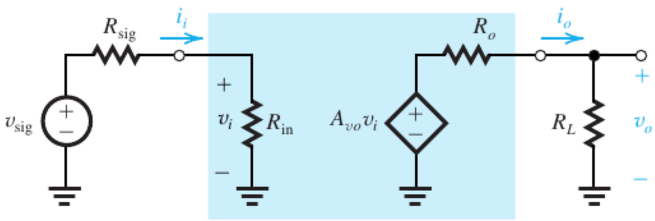


(b)



(c)

- the 3rd parameter that is essential to characterize an amplifier is the output resistance R_o
  - R_o is the resistance seen looking back into the amplfier output terminal with v_i set to zero
    - $= \Rightarrow R_o = \frac{v_x}{i_x}$
  - Note that the controlled source $A_{vo}v_i$ and the output resistance R_o
    - represent the Thevenin equivalent of the amplifier output circuit.



(b)

- by voltage divider
  - the output voltage v_o can be given as
    - $v_o = \frac{R_L}{R_L+R_o}\left(A_{vo}v_i\right) = \Rightarrow \frac{v_o}{v_i} = A_{vo}\frac{R_L}{R_L+R_o}$
  - thus the voltage gain of the amplifier is
    - $A_v = \frac{v_o}{v_i} = A_{vo}\frac{R_L}{R_L+R_o}$
  - the overall voltage gain is
    - $G_v = \frac{v_o}{v_{sig}} = A_{vo}\frac{R_L}{R_L+R_o}\frac{v_i}{v_{sig}}$
    - $G_v = A_{vo}\frac{R_L}{R_L+R_o}\frac{R_{in}}{R_{in}+R_{sig}} \because v_i = \frac{R_{in}}{R_{in}+R_{sig}}v_{sig}$
